

Conversion of Units Across Systems of Measure

Jonathan R. Bacolod

Measurement systems are standardized ways of expressing length, mass, capacity, and other quantities.

Common Measurement Systems

The two most common systems are the **International System of Units (SI)** and the **U.S. Customary System**.

Length Conversion Factors

$$1 \text{ inch} = 2.54 \text{ cm}$$

$$1 \text{ foot} = 30.48 \text{ cm}$$

$$1 \text{ yard} = 0.9144 \text{ m}$$

$$1 \text{ mile} = 1.609 \text{ km}$$

Mass Conversion Factors

$$1 \text{ pound} = 0.454 \text{ kg}$$

$$1 \text{ ounce} = 28.35 \text{ g}$$

Capacity Conversion Factors

$$1 \text{ gallon} = 3.785 \text{ L}$$

$$1 \text{ quart} = 0.946 \text{ L}$$

Converting Measurements

To convert measurements, **multiply** by the appropriate conversion factor.

$$12 \text{ inches} \times 2.54 = 30.48 \text{ cm}$$

Therefore, 12 inches = 30.48 centimeters.

Examples

Example 1

Convert 12 inches to centimeters.

$$12 \times 2.54 = 30.48$$

Therefore, 12 inches = 30.48 centimeters.

Example 2

Convert 5 feet to meters.

$$5 \times 30.48 = 152.4 \text{ cm}$$

$$152.4 \text{ cm} = 1.524 \text{ m}$$

Therefore, 5 feet = 1.524 meters.

Example 3

Convert 8 yards to meters.

$$8 \times 0.9144 = 7.3152$$

Therefore, 8 yards = 7.3152 meters.

Practice Exercises

Practice 1

Convert 12 inches to centimeters.

12 inches $\rightarrow$ centimeters

Practice 2

Convert 5 feet to meters.

5 feet $\rightarrow$ meters

Practice 3

Convert 8 yards to meters.

8 yards $\rightarrow$ meters

Practice 4

Convert 10 miles to kilometers.

10 miles $\rightarrow$ kilometers

Practice 5

Convert 15 pounds to kilograms.

15 pounds $\rightarrow$ kilograms

Practice 6

Convert 24 ounces to grams.

24 ounces $\rightarrow$ grams

Practice 7

Convert 3 gallons to liters.

3 gallons $\rightarrow$ liters

Practice 8

Convert 7 quarts to liters.

7 quarts $\rightarrow$ liters

Practice 9

Convert 36 inches to centimeters.

36 inches $\rightarrow$ centimeters

Practice 10

Convert 5 gallons to liters.

5 gallons $\rightarrow$ liters

Activities

Activity 1

Convert 18 inches to centimeters.

18 inches $\rightarrow$ centimeters

Activity 2

Convert 9 feet to meters.

9 feet $\rightarrow$ meters

Activity 3

Convert 15 yards to meters.

15 yards $\rightarrow$ meters

Activity 4

Convert 25 miles to kilometers.

25 miles $\rightarrow$ kilometers

Activity 5

Convert 20 pounds to kilograms.

20 pounds $\rightarrow$ kilograms

Activity 6

Convert 40 ounces to grams.

40 ounces $\rightarrow$ grams

Activity 7

Convert 6 gallons to liters.

6 gallons $\rightarrow$ liters

Activity 8

Convert 10 quarts to liters.

10 quarts $\rightarrow$ liters

Activity 9

Convert 48 inches to centimeters.

48 inches $\rightarrow$ centimeters

Activity 10

Convert 8 gallons to liters.

8 gallons $\rightarrow$ liters

Thank you!